Deep Learning for Natural Language Processing

Today's goal • 3/3 items

Module 1
Feedforward Neural Nets and Recurrent Neural Networks

Module 2
Sequence to Sequence Models, Attention, Transformers

Module 3
Transfer Learning

Pretraining, Finetuning, and Popular Pretrained Models

- Low-Resource Settings in NLP
  Video • 10 min
- Pretraining & Finetuning
  Video • 12 min
- Popular Pretrained Language Models: GPT&GPT-2
  Video • 9 min
- Popular Pretrained Masked Language Models: BERT&Co.
  Video • 9 min
- Domain Adaptation
  Video • 7 min
- Catastrophic Forgetting
  Video • 5 min
- Pretraining & Finetuning
  Practice Assignment • Grade: 80%

Multitask Training and Data Augmentation

- What Is Multitask Training?
  Video • 6 min
- Parameter Sharing for Multitask Training
  Video • 5 min
- Weighing of Losses
  Video • 6 min
- Task Combinations
  Video • 10 min
- Data Augmentation
  Video • 11 min
- Popular Multitask-Pretrained Models
  Video • 5 min
- Multitask Training and Data Augmentation
  Practice Assignment • Grade: 100%

Crosslingual Transfer and Multilingual Pretrained Models

- What Is Crosslingual Transfer?
  Video • 6 min
- Translation-based Crosslingual Transfer Approaches
  Video • 7 min



Programming assignment

Finetuning Machine Translation Models

Launch lab

What to expect

Due Nov 18, 11:59 PM IST

Unlimited attempts

You passed!
To pass you need at least 80/100 points.

100/100

Instructions | My submissions | Discussions

In the previous assignment, you learned the transformer architecture and trained one from scratch. In this assignment, you will finetune a pre-trained transformer model for machine translation. You will explore dataset preparation, tokenization, model training, and evaluation using the Hugging Face transformers library. The goal is to adapt an existing model to translate between a specific language pair, optimizing its performance on a given dataset. By the end of this assignment, you will have a deeper understanding of transfer learning and how to effectively finetune transformers for NLP tasks using Hugging Face.

For the purpose of this assignment, we will be using the MarianMT transformer model for the English to French translation task. The dataset of sentence pairs has been provided to you.

Report an issue

Go to next item →